

HINTS & SOLUTIONS

Q.1 (B)

$$\lim_{x \rightarrow 0^-} \left[\frac{4f(x) - 12}{\tan(2f(x) - 6)} \right] = \lim_{x \rightarrow 0^-} \left[\frac{2(2f(x) - 6)}{\tan(2f(x) - 6)} \right] = 1$$

Q.2 (A)

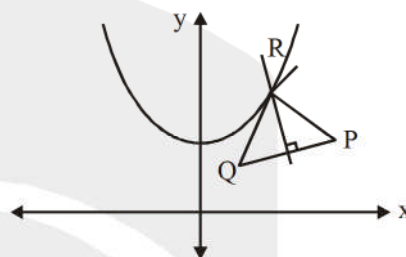
$$m_{PQ} = 2$$

$$\frac{dy}{dx} = 2x = 2$$

$$x = 1$$

$$\therefore R(1, 5)$$

$$A_{\min} = \frac{1}{2} \cdot \sqrt{5} \cdot \frac{4}{\sqrt{5}} = 2 \text{ Ans.}$$



Q.3 (C)

$$\lim_{h \rightarrow 0} \left(\frac{f(100 + 3h^3) - f(100 - 7h^3)}{h^3} \right) = \lim_{h \rightarrow 0} 10 \left(\frac{f(100 + 3h^3) - f(100 - 7h^3)}{10h^3} \right)$$

$$= 10 f'(100) = 100. \text{ Ans.}$$

Q.4 (D)

$$\text{Put } r = \frac{1}{x}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\cos(ax) - \cos(bx) \cdot \cos(cx)}{\left(\frac{\sin x(bx)}{bx} \right) \left(\frac{\sin(cx)}{cx} \right) \cdot bcx^2}$$

$$= \frac{1}{bc} \lim_{x \rightarrow 0} \frac{\cos(ax) - \cos(bx) \cdot \cos(cx)}{x^2} \quad (\text{use series expansion}) = \frac{b^2 + c^2 - a^2}{2bc}.]$$

Q.5 (D)

$$g'(x) = 2f(2f^2(x) + 2) \times f'(2f^2(x) + 2) \times 4f(x) \cdot f'(x)$$

$$g'(0) = 2f(2f^2(0) + 2) \times f'(2f^2(0) + 2) \times 4f(0) \cdot f'(0)$$

$$= 2f(4) \cdot f'(4) \cdot 4f(0) f'(0) = 2 \times 4(-1)(1) = -8. \text{ Ans.}$$

Q.6 (A)

$$\left. \begin{aligned} f'(1) &= -5, \quad f'(-1) = 5 \\ f'(3) &= -f'(1) = 5 \\ f'(-3) &= -f'(3) = -5 \end{aligned} \right\} \therefore f'(k) = \begin{cases} 5, & \text{if } k = 4r-1, \quad r \in I \\ -5, & \text{if } k = 4r+1, \quad r \in I \end{cases}$$

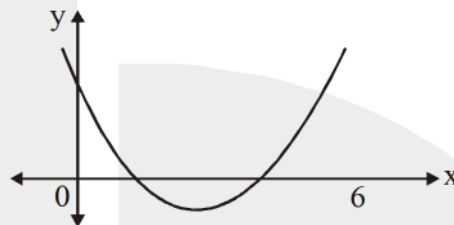
$$\sum_{r=0}^{100} (-1)^r f'(r) = f'(0) - f'(1) + f'(2) - f'(3) + \dots - f'(99) + f'(100) = 0. \quad]$$

Q.7 (C)

Both roots are distinct and lies in (0, 6)

$$\left. \begin{aligned} D &> 0 \\ f(0) &> 0 \\ f(6) &> 0 \end{aligned} \right\} \Rightarrow k \in (2\sqrt{6}, 7)$$

$\therefore k_{\max} = 6$. **Ans.**



Note: From the range of k , those values of k will be rejected where $x^2 - kx + 6$ vanishes at $x = \frac{\pi}{6}$ or

$$\frac{5\pi}{6}.]$$

Q.8 (D)

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f\left(\frac{2x+2h}{2}\right) - f\left(\frac{2x+0}{2}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\left(\frac{f(2x) + f(2h)}{2}\right) - \left(\frac{f(2x) + f(0)}{2}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(2h) - f(0)}{2h} \end{aligned}$$

$$f'(x) = f'(0)$$

$$\therefore f(x) = f'(0) x + \lambda$$

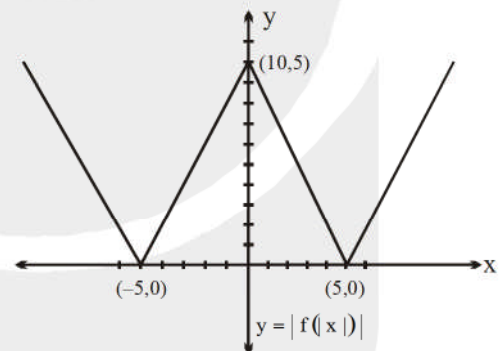
$$\text{As } f(0) = 5$$

$$\Rightarrow f(x) = f'(0) x + 5$$

$$\text{Also } f'(x) = f'(0) \Rightarrow f'(5) = f'(0) = -1$$

$$\Rightarrow f'(0) = 1$$

$$\text{Hence, } f(x) = 5 - x. \quad \text{Ans.}$$



Q.9 (B)

$$\therefore 6 \leq f(x) \leq 7.5 \quad \text{and} \quad -5 \leq 3 \sin \alpha + 4 \cos \alpha \leq 5$$

$$\Rightarrow 1 \leq 3 \sin \alpha + 4 \cos \alpha + f(x) \leq 12.5$$

$$\Rightarrow \frac{1}{\lambda} \leq \frac{3 \sin \alpha + 4 \cos \alpha + f(x)}{\lambda} \leq \frac{12.5}{\lambda}$$

Clearly $h(x)$ will be continuous if $\left[\frac{1}{\lambda}, \frac{12.5}{\lambda}\right]$ does not contain any integer.

$$\therefore \lambda > 12.5 \Rightarrow \lambda \geq 13$$

$\therefore$ number of values of $\lambda = 8$. **Ans.**

Q.10 (D)

$$f(x) = \left[\underbrace{\frac{a}{3}x + b}_E \right] \text{ is continuous in } [0, 2]$$

$$x = 0 \quad E = b \geq 0$$

$$x = 2 \quad E = \frac{2a}{3} + b \geq 0 \quad (E \text{ is increasing function or constant function})$$

$$\frac{2a}{3} + b - b < 1$$

$$a < \frac{3}{2} \Rightarrow a = 0, 1$$

where $a = 0$, then b has 11 values

where $a = 1$, then b has 11 values

$\therefore$ (22 ordered pairs).

Paragraph for question nos. 11 & 12

Q.11 (D) & Q.12 (B)

(i) $f(1, y) = \frac{4\sin^2 y}{y^2}$ (putting $x = 1$ & dividing Nr & Dr by 2^n & taking $n \rightarrow \infty$)

$\therefore$ continuous for all $y \in \mathbb{R}$

$$\therefore f(1, 0) = \lim_{y \rightarrow 0} \frac{4\sin^2 y}{y^2} = 4$$

(ii) $\lim_{x \rightarrow 0} f(x, 0) = \lim_{\substack{n \rightarrow \infty \\ x \rightarrow 0}} \frac{2^{n+2} \sin^2(x-1) + \tan^2 x + \frac{2}{\pi} x^2 \tan^{-1} x}{2^{nx} (x-1)^2 + \lambda x^4 \operatorname{cosec}^2 1}$

$$= \lim_{\substack{n \rightarrow \infty \\ x \rightarrow 0^+}} \frac{\sin^2(x-1) + \frac{\tan^2 x + \frac{2}{\pi} x^2 \tan^{-1} nx}{2^{nx+2}}}{\frac{(x-1)^2}{2^2} + \frac{\lambda x^4 \operatorname{cosec}^2 1}{2^{nx+2}}} = 4 \sin^2 1.$$

$$\lim_{\substack{n \rightarrow \infty \\ x \rightarrow 0^-}} \frac{0 + \tan^2 x - x^2}{\lambda x^4 \operatorname{cosec}^2 1}$$

$$\frac{\sin^2 1}{\lambda} \left(\frac{2}{3} \right)$$



for existence of limit

$$\Rightarrow 4 = \frac{2}{3\lambda} \Rightarrow \lambda = \frac{1}{6} \text{ . Ans.}$$

Q.13 (ABC)

$$f(x) = \cos 2x \text{ and } g(x) = e^x$$

Q.14 (BCD)

$$y = \sqrt[3]{6xy^2 - 11x^2y + 6x^3}$$

$$y^3 = 6xy^2 - 11x^2y + 6x^3$$

$$y^3 - 6xy^2 + 11x^2y - 6x^3 = 0$$

$$\Rightarrow (y-x)(y-2x)(y-3x) = 0$$

$$\Rightarrow y = x, y = 2x, y = 3x$$

$$\therefore \frac{dy}{dx} = 1, 2, 3$$

Q.15 (C)

$$L = \lim_{x \rightarrow 0} \frac{a \left(x - \frac{x^3}{3!} + \dots \right) + bx \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots \right) + 3x^2}{\left(\frac{\sin x - x}{x^3} - \frac{(x - \tan x)}{x^3} \right) x^3}$$

$$L = 6 \lim_{x \rightarrow 0} \frac{x(a+b) + x^2(b+3) + x^3 \left(\frac{b}{2} - \frac{a}{6} \right) + \dots}{x^3} \Rightarrow L = 6 \left(\frac{b}{2} - \frac{a}{6} \right)$$

$$\Rightarrow a + b = 0 \text{ and } b + 3 = 0 \Rightarrow b = -3. \text{ Also } a = 3$$

$$L = 6 \left(-\frac{3}{2} - \frac{3}{6} \right) = -12. \text{ Hence, } \left(\frac{b-L}{a} \right) = \frac{-3 - (-12)}{3} = \frac{9}{3} = 3 \text{ Ans.}$$

Q.16 (D)

$$y = f(x) = \frac{-2(x^4 + x^2 + 1) + 3x^2}{x^4 + x^2 + 1}$$

$$f(x) = \frac{3x^2}{x^4 + x^2 + 1} - 2$$

Range of $f(x)$ is $[-2, -1]$ (-2 occur at $x = 0$ & -1 occur at $x = 1$)

(P) $M = -1$; $|4M| = 4$

(Q) $[f(x)]$ is discontinuous when $f(x) = -1$

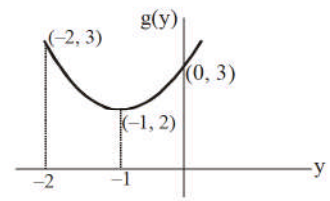
$$\frac{-2x^4 + x^2 - 2}{x^4 + x^2 + 1} = -1$$

$$x = \pm 1$$

(R) Let $f(x) = y$ $y \in [-2, -1]$

$$\text{now } g(y) = y^2 + 2y + 3$$

$$2 \leq g(y) \leq 3$$



Hence $[g \circ f(x)] = 2 \& 3$

Hence, it is discontinuous at $f(x) = -2$

$\therefore$ Number of points of discontinuity is one.

(S) $\text{sgn}(g(x) + 2 - p^2)$ is discontinuous at exactly one point

$$g(x) + 2 - p^2 = x^2 + 2x + 3 + 2 - p^2 = 0$$

$$x^2 + 2x + 5 - p^2 = 0$$

$$D = 0$$

$$p = \pm 2$$

$$\text{Product} = -4.$$

Q.17 [4]

$$f(x) = \underbrace{\cos(2x-1)\frac{\pi}{2}}_{\substack{\text{differentiable at} \\ x = -1, 5 \text{ and} \\ \text{value} = 0 \text{ at} \\ \text{these points}}} \cdot \underbrace{|(x+1)(x-5)|}_{\substack{\text{continuous} \\ \text{at } x = -1, 5}} + \underbrace{x^2|x|(x^2+1)}_{\substack{\text{differentiable} \\ \text{at } x = 0}} + \underbrace{x}_{\substack{\text{differentiable} \\ \text{at } x = 0 \text{ and} \\ \text{has value} = 0 \\ \text{at } x = 0}} \cdot \underbrace{e^{|x|}}_{\substack{\text{continuous} \\ \text{at } x = 0}}$$

differentiable on $\mathbb{R}$

$$\Rightarrow n_1 = 0$$

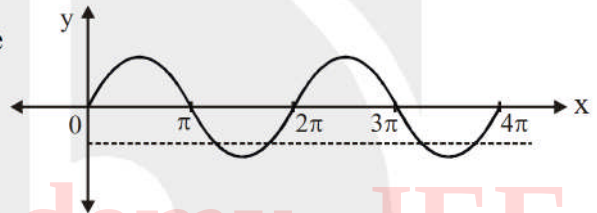
$g(x) = \text{sgn}(\sin^2 x - \sin x - 1)$ is discontinuous where

$$\therefore (\sin^2 x - \sin x - 1) = 0$$

$$\Rightarrow \sin x = \frac{1 \pm \sqrt{5}}{2} \Rightarrow \sin x = \frac{1 - \sqrt{5}}{2}$$

i.e. it changes sign 4 times.

$$\Rightarrow n_2 = 4 \Rightarrow n_1 + n_2 = 4]$$



Q.18 [2]

$$\therefore \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 8x + 3} - \sqrt{x^2 + 4x + 2} \right) = \lim_{x \rightarrow \infty} \frac{(x^2 + 8x + 3) - (x^2 + 4x + 2)}{\left(\sqrt{x^2 + 8x + 3} + \sqrt{x^2 + 4x + 2} \right)}$$

$$= \lim_{x \rightarrow \infty} \left(\frac{4 + \frac{1}{x}}{\sqrt{1 + \frac{8}{x} + \frac{3}{x^2}} + \sqrt{1 + \frac{4}{x} + \frac{2}{x^2}}} \right) = \frac{4}{1+1} = 2$$

$$\therefore \tan^2 \alpha + \frac{1}{\tan^2 \alpha} = 2 \Rightarrow \tan^2 \alpha = 1 \Rightarrow \sin^2 \alpha = \cos^2 \alpha = \frac{1}{2}.$$

$$\therefore \sum_{n=1}^{\infty} (\sin^{2n} \alpha + \cos^{2n} \alpha)$$

$$\sum_{n=1}^{\infty} \left(\frac{1}{2} \right)^n + \left(\frac{1}{2} \right)^n = 2 \sum_{n=1}^{\infty} \frac{1}{2^n} = 2 \left(\frac{1}{2} + \frac{1}{2^2} + \dots \right) = 2 \text{ Ans.}$$

Q.19 [3]

$$f^2(x+y) = f^2(x) + f^2(y) - 2\sqrt{x} - 2\sqrt{y} + 2\sqrt{x+y} - 1$$

Partially differentiating w.r.t. x

$$2f(x+y)f'(x+y) = 2f(x)f'(x) - \frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x+y}}$$

Putting $x = 1$

$$2f(1+y)f'(1+y) = 2f(1)f'(1) - 1 + \frac{1}{\sqrt{1+y}}$$

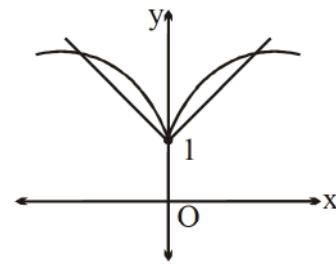
$$2f(1+y)f'(1+y) = 2 \cdot 2 \cdot \frac{1}{2} - 1 + \frac{1}{\sqrt{1+y}}$$

$$2f(1+y)f'(1+y) = 1 + \frac{1}{\sqrt{1+y}}$$

$$2f(x)f'(x) = 1 + \frac{1}{\sqrt{x}} \Rightarrow (f(x))^2 = x + 2\sqrt{x} + C$$

Putting $x = 1, C = 1$

$$\Rightarrow f(x) = \sqrt{x} + 1.$$



Q.20 [0]

$$(x+1)^3 + (y+1)^3 = 16$$

$$y = -1 + (16 - (x+1)^3)^{1/3} = f(x)$$

$$x = -1 + (16 - (y+1)^3)^{1/3}$$

$$\therefore f^{-1}(x) = -1 + (16 - (x+1)^3)^{1/3} = g(x)$$

$$\Rightarrow f(x) = g(x)$$

$$f(-1 + \sqrt[3]{15}) = g(-1 + \sqrt[3]{15}) = 0$$

$$(f(x+g(x)) \cdot g(x+f(x))) = (f(x+f(x)))^2$$

$$\frac{d}{dx}(f(x+f(x)))^2 = 2f(x+f(x)) \cdot f'(x+f(x)) \cdot (x+f'(x))$$

$$\frac{d}{dx}(f(x+f(x)))^2 \text{ at } x = -1 + \sqrt[3]{15} = 0$$